

The Hidden Cost of "Free": Open Educational Resources, Academic Rigor, and the Undermining of Higher Education Quality

The Issue

In light of the critical challenges surrounding student debt and the soaring costs of textbooks—often reaching \$1,000 to \$1,500 per semester—universities and colleges must confidently embrace the adoption of Open Educational Resources (OER). These invaluable resources, which include free, openly licensed textbooks and online course materials, are essential for easing the financial strain on students. This bold shift directly addresses the pressing issue of textbook affordability, especially given that traditional commercial textbooks typically range from \$200 to \$400 each, with new editions released every 2–3 years, quickly rendering used copies obsolete.

While promoting OER is a commendable initiative, it is equally important to scrutinize the implications for educational quality. Many OER materials offer substantial value; however, some may not align with the rigorous standards expected in academic environments. There are instances where these resources can lack comprehensive coverage, fail to thoroughly explain complex concepts, or provide inadequate practical examples and problem sets that are critical for skill development. Furthermore, the quality of illustrations, diagrams, and multimedia elements may not always meet the standards found in commercial materials. The absence of supplementary resources—such as test banks and interactive tools—can limit the effectiveness of teaching and learning experiences, with quality control processes often less stringent than those for traditionally published content.

We must acknowledge the risks that come with relying solely on OER. Students utilizing these resources might not achieve the depth of understanding necessary for success in advanced studies and professional settings. This gap can result in graduates feeling ill-equipped for their future careers and academic challenges. Therefore, it is essential to champion resources that not only offer short-term cost savings but also significantly enhance educational quality and foster long-term student success.

Universities have a pivotal role in strategically balancing cost-saving measures with the unwavering commitment to maintain high academic standards. Open discussions about the selection of OER and their impact on education are critical in uncovering opportunities for innovation in teaching methodologies. By prioritizing both affordability and quality, we can ensure that students receive a superior education, fully preparing them for their professional journeys. This approach not only drives improved student outcomes but also reinforces the institution's dedication to providing exceptional education while tackling rising tuition costs. Together, we are poised to navigate this transformative era in educational resources with confidence and a clear focus on the shared goal of student success.

Why It Matters

The Textbook Affordability Crisis and Institutional Response: Real Problem, Problematic Solution

The Crisis Is Real:

- The high cost of commercial textbooks has become a serious problem, making it harder for students to succeed and causing frustration. According to the U.S. Government Accountability Office (2013), textbook prices rose by 82% from 2002 to 2012, much faster than the 28% inflation rate during that time. The College Board (2021) reports that students at

public four-year colleges spend about \$1,240 each year on books and supplies, while community college students spend around \$1,460. Some textbooks, especially in STEM subjects, can cost \$200 to \$400 or more (Senack, 2016).

- Research shows that publishers use several unfair practices. They often release new editions every two or three years with few real changes, which makes it hard for students to buy used books (Hilton et al., 2013). Publishers also bundle textbooks with costly, single-use online access codes that cannot be resold (Florida Virtual Campus, 2016). In addition, they charge U.S. students much more than international students for the same content (Jhangiani & Jhangiani, 2017).
- These high costs make things difficult for students. In a survey of more than 22,000 Florida college students, the Florida Virtual Campus (2016) found that 66.6% did not buy a required textbook because it was too expensive. As a result, 37.6% got a poor grade, 19.8% failed a course, and 47.6% took fewer classes because of textbook costs. Colvard et al. (2018) also found that textbook costs hurt student success, especially for Pell Grant recipients, underrepresented minorities, and part-time students.

But Why Are Universities REALLY Pushing OER?

- Universities often gain public relations and political benefits from Open Educational Resources (OER) initiatives, but these efforts rarely tackle the real reasons behind rising college costs. Henson (2017) studied OER policies at 50 institutions and found that most schools were motivated by the desire for positive media attention and marketing, deflecting criticism about tuition hikes by pointing to textbook savings, responding to political pressure to address college costs without cutting tuition or administrative spending, attracting students who are concerned about costs, and showing innovation and a focus on students to accreditors and stakeholders.
- Data shows that OER initiatives are often more about appearances than real solutions. Textbooks make up only about 2-4% of the total cost to attend college, while tuition and fees account for 50-70% (College Board, 2021). Even so, schools often claim that OER saves students \$500-\$1,000 a year, while at the same time raising tuition by \$1,000-\$3,000 each year. This means students end up paying more overall, even with these supposed savings. According to SHEEO (2020), from 2015 to 2020, when OER use became common, average tuition at public four-year colleges rose by 20-35% after inflation adjustment, far more than any savings from textbooks.
- OER initiatives enable institutions to present themselves as cost-conscious without undertaking challenging budgetary reforms. Instead of reducing administrative positions—which have increased 60% faster than enrollment between 2000 and 2020 (Ginsberg, 2011)—decreasing executive compensation, limiting expenditures on amenities and facilities, or curbing spending on athletics and marketing, institutions often mandate "free" textbooks. This approach shifts costs to faculty, in the form of uncompensated labor, and to students, through potentially diminished learning experiences. Targeting textbook costs represents a politically expedient strategy that circumvents more substantive discussions regarding institutional priorities.
- Belikov and Bodily (2016) found that institutional motivations often diverged from publicly stated values. While OER initiatives were publicly framed in terms of "student success" and "access," internal documents and administrator interviews indicated that actual priorities included cost reduction, positive publicity, and competitive advantage in enrollment. Furthermore, only 23% of institutions with OER mandates monitored student learning outcomes to ensure educational quality.

The OER Landscape: Major Repositories and Adoption Rates

Major OER Sources and Repositories:

- OpenStax, headquartered at Rice University, is a prominent provider of open educational resources (OER) that offers peer-reviewed, professionally developed textbooks primarily for large introductory courses. By 2022, more than 3.6 million students at over 2,500 institutions had utilized OpenStax materials in 14,000 courses, resulting in an estimated \$1.5 billion in textbook cost savings (OpenStax, 2022). OpenStax publishes approximately 60 textbooks spanning

mathematics, science, social sciences, humanities, and business. Each textbook undergoes expert review, professional editing and design, and aligns with conventional textbook standards.

- The BCcampus Open Collection, based in British Columbia, Canada, offers over 300 open textbooks across a range of subjects, primarily serving Canadian institutions. Since 2012, these textbooks have been adopted in more than 1,000 courses, resulting in student savings of approximately \$20 million CAD (BCcampus, 2021). The collection comprises both BCcampus-funded textbooks and selected external resources. BCcampus provides detailed reviews using standardized rubrics to assist faculty in evaluating quality, a feature that distinguishes it from many other repositories.
- The Open Textbook Library at the University of Minnesota catalogs over 950 open textbooks from diverse sources within a searchable, subject-based database. By 2023, these textbooks had been adopted in thousands of courses, although precise usage data is not systematically tracked. Faculty reviews and ratings are available; however, the level of detail and consistency varies. Because the library primarily aggregates rather than produces textbooks, quality control depends on the standards of the original sources.
- LibreTexts is a large collaborative project offering over 150,000 pages of open educational content across 13 subject libraries, including Chemistry, Mathematics, Biology, Physics, Engineering, Medicine, Humanities, and Social Sciences. In 2021, LibreTexts materials were accessed more than 1 billion times, used by over 5 million students monthly, and adopted in more than 12,000 courses at 1,500 institutions (Delmas et al., 2022). Faculty can remix, adapt, and add content, making LibreTexts a "living library." The Chemistry section is particularly extensive, with much of its content originating from the UC Davis ChemWiki project. Due to ongoing updates, there are concerns about quality control and version consistency.
- The American Institute of Mathematics (AIM) Open Textbook Initiative specializes in mathematics textbooks, approving only those deemed "high quality and open source." As of 2023, AIM lists approximately 90 approved textbooks covering topics ranging from basic to advanced undergraduate mathematics. Each book undergoes review for content quality, mathematical accuracy, teaching effectiveness, and open licensing, providing a level of quality assurance uncommon among other repositories.
- Other significant OER repositories include MERLOT, with over 40,000 peer-reviewed resources; OER Commons, with more than 100,000 resources; and MIT OpenCourseWare, which provides materials from over 2,400 MIT courses. Additional important resources comprise OpenLearn from the Open University UK, Saylor Academy's collection of over 100 open courses, and subject-specific platforms such as Project Gutenberg for literature, PhET Interactive Simulations for science, and Khan Academy for multiple subjects.

OER Adoption Statistics and Trends:

- It is hard to determine exact adoption rates because institutions track and report data differently. Still, Seaman and Seaman (2021) found that about 8-10% of higher education faculty in the United States have used OER as the main textbook for at least one course, up from about 3% in 2015-2016. This means OER adoption has more than doubled over the past five to six years, but it still accounts for only a small share of all courses.
- Adoption rates vary widely by institution type and state policies. Seaman and Seaman (2021) found that community colleges have higher OER adoption (12-15% of courses) than four-year schools (6-9%). Public institutions use OER more than private ones (10% compared to 5%). States with OER funding programs, like California, Georgia, Washington, and Virginia, have adoption rates two to three times higher than states without such support. Large introductory courses also use OER more (15-25%) than upper-division or specialized courses (3-7%).
- Lower-division courses use OER much more than upper-division courses. Exact numbers depend on the institution and subject, but research shows that introductory general education courses like English Composition, College Algebra, Introduction to Psychology, Principles of Economics, General Biology, and General Chemistry have OER adoption rates of 15-30% at schools with OER programs. Upper-division major courses have adoption rates of 5-10%, while specialized,

advanced, or graduate courses have very low adoption (2-5% or less), mainly because there are not enough high-quality OER materials for these topics (Seaman & Seaman, 2021; Croteau, 2017).

- State-level initiatives dramatically increase adoption. Florida Virtual Campus (2021) reported that Florida's OER initiatives resulted in over 6,000 course sections adopting OER materials across Florida's public colleges and universities, representing approximately 12-15% of high-enrollment general education courses. Similarly, Affordable Learning Georgia (Croteau, 2017) documented over 1,000 course sections at Georgia institutions adopting OER, with a concentration in lower-division general education courses at University System of Georgia schools.
- California's ambitious OER initiatives show mixed results. California's large OER programs have had mixed results. The California OER Council and the California Community Colleges Chancellor's Office have spent millions on OER development and adoption. Ozdemir and Hendricks (2017) found that about 10-15% of courses at California community colleges use OER, with higher rates (20-30%) in high-enrollment courses like English, Math, and Psychology. Other subjects and upper-division courses have lower rates (5% or less). In 2020, the California State University system reported OER use in over 2,000 course sections serving more than 76,000 students each year, but this is less than 5% of all course sections across its 23 campuses.⁷) and Seaman and Seaman (2021) found that: mathematics and statistics show relatively high OER adoption (12-18% of introductory courses) due to availability of quality OER like OpenStax and AIM-approved textbooks, sciences show moderate adoption (10-15% in introductory Biology, Chemistry, Physics) primarily through OpenStax materials, social sciences vary widely (Psychology 15-20% adoption, Sociology 10-15%, Economics 8-12%, Political Science 5-8%), humanities show low adoption (5-10%) except for some literature courses using public domain texts, and professional programs show very low adoption (3-5%) in nursing, business, engineering due to lack of current, specialized OER and accreditation requirements.
- OER adoption is growing quickly, but it started from a low point. Seaman and Seaman (2021) report that OER use increased by 150-200% from 2016 to 2021, rising from about 3-4% to 8-10% of courses. This is a significant percentage increase, but it still reaches only a minority of students. Without major new funding or requirements, OER adoption may level off at 15-20% of courses, mostly in lower-division general education, instead of becoming the norm.
- The impact of the COVID-19 pandemic on OER adoption is unclear. The effect of the COVID-19 pandemic on OER adoption is still uncertain. Some thought that moving to remote learning would accelerate OER adoption due to concerns about access and cost, but Seaman and Seaman (2021) found mixed results. Some schools saw increased interest in OER during the pandemic, but many faculty, busy with emergency remote teaching, kept using familiar materials rather than switching to new ones. The long-term effects of the pandemic on OER adoption remain unclear.

Institutional Mandates and Incentives:

- A growing number of institutions have implemented OER mandates, strong incentives, or degree pathways. Ozdemir and Hendricks (2017) documented that, as of 2017, at least 40 U.S. institutions had established "OER degrees" or "Z-degrees" (zero-cost textbook degrees), in which all or nearly all required courses use OER materials. By 2021, this number had grown to 100+ institutions (SPARC, 2021), though these programs typically serve small numbers of students and concentrate in specific majors (often general studies, liberal arts, or social sciences) where OER availability is better.
- State-level mandates and funding create pressure for adoption. At least 15 states have enacted legislation or funding initiatives that promote or require OER adoption in public institutions (SPARC, 2021). These range from funding grants for OER development or adoption (California, Georgia, Washington, Virginia), to requirements that institutions track and report OER usage (Florida, Texas), to mandates that OER be considered for high-enrollment courses (Oregon, Maryland), to the creation of state OER libraries and repositories. Belikov and Bodily (2016) found that faculty in states with such policies feel increased pressure to adopt OER regardless of quality or pedagogical fit.

What Statistics Reveal:

- Despite growth and investment, OER remains minority practice affecting 8-10% of courses overall, concentrated heavily in: lower-division general education courses (15-30% adoption), introductory STEM and social sciences where quality OER exists (10-20% adoption), community colleges rather than four-year institutions (12-15% vs. 6-9%), institutions in states with OER funding and mandates (15-20% vs. 5-7%), and specific institutions with aggressive OER initiatives (some reporting 25-40% of courses using OER).
- The vast majority of courses—90%+ overall, and 95%+ of upper-division courses—still use commercial textbooks or other non-OER materials. This suggests either that quality OER doesn't exist for most courses, especially specialized upper-division courses; that faculty judge available OER to be inadequate for their courses; or that insufficient incentives/pressure exist to overcome faculty concerns about OER adoption. The concentration of OER in lower-division general education courses where commercial textbooks are most expensive and standardized suggests strategic targeting by institutions—addressing the most visible, politically sensitive textbook costs rather than comprehensive OER transformation.
- If adoption continues at current growth rates (doubling every 5-6 years), OER might reach 15-20% of courses by 2025-2027, 25-30% by 2030. However, this assumes continued funding and institutional support, the development of high-quality OER for currently underserved subjects and upper-division courses, and faculty acceptance of OER quality and pedagogical fit. Without addressing quality concerns documented in research (Bliss et al., 2013; Belikov & Bodily, 2016), adoption may plateau well below majority practice

The Quality Problem: Research Evidence of OER Limitations and Gaps

OER Quality Is Highly Variable and Often Inadequate:

- Research on OER quality reveals significant concerns. Delimont et al. (2016) surveyed 108 educators and found that perceived barriers to OER development included: lack of time (91%), lack of institutional support (58%), lack of technical expertise (53%), concerns about quality control (47%), and lack of funding (45%). These barriers suggest that much OER is created under less-than-ideal conditions.
- Quality assessments of OER reveal substantial gaps. Bliss et al. (2013) evaluated the quality of OER using the LORI (Learning Object Review Instrument) framework and found significant variability in quality, with many OER scoring poorly on content quality, learning goal alignment, feedback and adaptation, and motivation. The study noted that "the quality of OER is highly variable" and recommended careful evaluation before adoption.
- Faculty report concerns about the quality and completeness of OER. Allen and Seaman (2014) found that only 11.3% of teaching faculty were aware of and using OER, and Belikov and Bodily (2016) found that faculty concerns about OER quality were the primary barrier to adoption, with faculty reporting that OER often lacked comprehensiveness, had gaps in content coverage, and required significant adaptation and supplementation.
- In STEM fields, particularly, OER problem sets are often inadequate. Winitzky-Stephens and Pickavance (2017) found that faculty in mathematics and sciences reported that available OER lacked sufficient practice problems, worked examples, and variety of problems—essential components for developing quantitative and analytical skills. Faculty reported spending 10-20 additional hours per course creating supplementary problems to address OER gaps.

Peer Review and Vetting Gaps:

- Traditional textbooks undergo extensive peer review and editorial processes. Pawlyshyn et al. (2013) document that commercial textbook development involves multiple rounds of expert review, developmental editing, fact-checking, student and instructor pilot testing, and typically 2-4 years of development with teams of 10-30 experts. Publishers invest \$500,000-\$1,000,000+ in major textbook development because their business depends on quality.
- OER lacks comparable systematic vetting. While some OER repositories like MERLOT and OpenStax have review processes, much OER is self-published without external review. Jung et al. (2017) found that only 23% of OER in major

repositories had undergone formal peer review, compared to nearly 100% of commercial textbooks. Clements and Pawlowski (2012) identified quality assurance as one of the most significant challenges in OER, noting that "lack of quality assurance mechanisms is a major impediment to OER adoption."

- Faculty lack time and sometimes expertise to thoroughly vet OER quality. Belikov and Bodily (2016) found that faculty reported needing 20-40 hours to adequately evaluate OER before adoption—time they often don't have—and that evaluation was particularly challenging outside their subspecialty areas.

Currency and Updating Problems:

- Keeping content current is essential, particularly in rapidly evolving fields. Research shows that outdated materials harm learning outcomes, especially in STEM, technology, healthcare, and social sciences (Fischer et al., 2015).
- OER updating is inconsistent and often abandoned. Pitt (2015) found that many OER materials become "orphaned" when their original authors leave institutions or stop maintaining them, and that 40-60% of OER in major repositories were more than 5 years old and lacked updates. Unlike commercial publishers, whose business models ensure ongoing updates, OER lacks sustainable maintenance structures.
- Faculty report outdated OER as significant concern. Belikov and Bodily (2016) found that 38% of faculty cited outdated content as a barrier to OER adoption, and Delimont et al. (2016) found that faculty creators often lacked time or incentive to maintain OER after initial creation.

Supplementary Materials Gap:

- Commercial textbooks include extensive ancillary materials that enhance teaching and learning. Pawlyshyn et al. (2013) documented that typical commercial textbook packages include instructor manuals, test banks with 500-2,000 questions, PowerPoint slides, solutions manuals, student study guides, online homework systems, and multimedia resources—representing millions of dollars in development investment.
- Most OER lacks these materials or has minimal versions. Bliss et al. (2013) found that only 15% of OER had comprehensive instructor resources, and fewer than 10% had test banks or online homework systems. This creates significant additional workload for faculty and reduces teaching quality.
- Faculty report a gap in supplementary materials as a major barrier. Jung et al. (2017) found that 67% of faculty cited lack of ancillary materials as barrier to OER adoption, and Winitzky-Stephens and Pickavance (2017) found that faculty adopting OER spent an average of 15-20 hours creating supplementary materials—time that could have been spent on student interaction and mentoring.

Faculty Perspectives: Benefits, Burdens, and Mixed Feelings About OER

Why Some Faculty Support OER:

- Some faculty are genuinely committed to educational access and equity, viewing OER as moral imperative. Kimmons (2015) found that faculty who voluntarily adopt OER cite: a philosophical commitment to open education and knowledge as a public good; a desire to reduce financial barriers for students (particularly low-income students); frustration with commercial publishers' exploitation and profiteering; and a commitment to educational justice and equity. These faculty often invest enormous unpaid time creating high-quality OER because they believe in the mission.
- Some faculty receive tangible benefits or recognition for OER work. Jung et al. (2017) documented that institutions sometimes provide: small stipends or course releases for OER creation (\$500-\$5,000 one-time payments or single course release), grant funding for OER development projects, professional recognition and awards (though rarely counting toward tenure/promotion), conference travel funding to present on OER work, and visibility/reputation within open education community. However, these benefits are inconsistent and often inadequate compensation for time invested.

- Faculty gain autonomy and ability to customize materials. Rather than being constrained by commercial textbook structure, OER allows faculty to: remix and adapt content to fit specific learning objectives and student needs, incorporate diverse voices and current events easily, create culturally responsive materials reflecting their students, update content immediately when new developments occur, and own their intellectual work rather than assigning rights to publishers (Petrides et al., 2011).

But Most Faculty Feel Pressured, Burdened, and Concerned:

- The majority of faculty report feeling pressured or coerced into OER adoption against their professional judgment. Belikov and Bodily (2016) found that: 63% of faculty at institutions with OER policies felt pressure to adopt OER regardless of quality or fit, 71% reported inadequate institutional support for OER development and implementation, 58% said OER adoption was "strongly encouraged" in ways that felt like soft mandates, 47% expressed concerns that OER quality was insufficient for their courses, and 82% said they would not have chosen OER without institutional pressure. Faculty describe feeling that their professional autonomy and expertise are being disregarded.
- Faculty express frustration that OER mandates shift costs from institutions to their unpaid labor. Delimont et al. (2016) documented that faculty creating or adapting OER spend 20-60 additional hours per course—time that is: uncompensated (no extra pay, course release typically inadequate), invisible to students and administrators, taken from research, advising, family time, and other responsibilities, and undervalued (not counted toward tenure/promotion in most cases). Petrides et al. (2011) calculated this represents \$15,000-\$30,000 in faculty labor per course when valued at academic hourly rates. Faculty resent being told to absorb costs so institutions can claim affordability while raising tuition.
- Faculty worry about educational quality and student learning outcomes. Surveys consistently show faculty concerns that: OER materials are less comprehensive and rigorous than commercial alternatives (Bliss et al., 2013), students are not learning as deeply or retaining material as well (Croteau, 2017), gaps in OER coverage require extensive supplementation (Winitzky-Stephens & Pickavance, 2017), and lack of ancillary materials (test banks, online homework, multimedia) makes teaching less effective (Jung et al., 2017).
- Many faculty lack time, expertise, or resources to create quality OER. Allen and Seaman (2014) found that: 71% of faculty cited lack of time as barrier to OER creation, 53% cited lack of technical expertise (particularly for multimedia and interactive content), 58% cited lack of institutional support (IT, instructional design, editorial assistance), 45% cited lack of funding for development costs, and 67% said they didn't know where to find quality OER or how to evaluate it.

Adjunct Faculty Are Particularly Harmed:

OER mandates disproportionately burden already-exploited adjunct faculty. As documented in "The Precarious Professor" section, 70%+ of college courses are taught by adjuncts earning poverty wages (\$2,000-\$5,000 per course, no benefits). These faculty: cannot afford 20-60 unpaid hours adapting OER when teaching at multiple institutions for survival, need ready-to-use materials with test banks and supplements to manage crushing workloads, have no job security to risk resisting institutional mandates, and receive no stipends or support for OER work. Mandating that the most exploited faculty absorb additional unpaid labor is particularly egregious.

Impact on Student Learning, Enrollment, and the Hidden Costs of "Free"

Learning Outcomes: Mixed Evidence with Methodological Limitations:

- Research on OER learning outcomes shows mostly equivalent performance on basic measures, but severe methodological limitations prevent confident conclusions. Hilton's (2016) comprehensive review of OER efficacy studies found that most showed no significant difference in learning outcomes between OER and commercial textbooks, but noted severe methodological problems: small sample sizes, lack of randomization, publication bias (negative results

unpublished), outcome measures limited to course grades and basic recall, higher-order thinking rarely assessed, and lack of long-term follow-up on retention and application.

- Studies showing equivalent outcomes often measure only surface-level learning. Colvard et al. (2018) found equivalent grades with OER but acknowledged that course grades measure "successful navigation of course requirements" not necessarily depth of learning or long-term retention. Clinton and Khan (2019) note that most OER efficacy research uses course performance as outcome measure, which may not capture conceptual understanding or transfer.
- Some evidence suggests OER may disadvantage students on higher-order outcomes. Croteau (2017) found that while students using OER performed equivalently on multiple-choice exams (recall and recognition), they scored lower on essay exams and application problems requiring deeper understanding. Similarly, Grewe and Davis (2017) found that students using OER in statistics courses performed worse on conceptual understanding items despite equivalent performance on computational problems.
- Students report dissatisfaction with OER quality despite cost savings. Jhangiani and Jhangiani (2017) found that while students appreciated cost savings, 42% rated OER quality lower than commercial textbooks, and 38% reported needing to seek supplementary resources. Florida Virtual Campus (2016) found that 35% of students using OER rated them as lower quality than commercial options.
- Long-term impacts on retention, transfer, and advanced coursework are completely unstudied. No published research examines whether students who use OER in foundational courses perform worse in subsequent advanced courses due to knowledge gaps—a critical question for understanding the true educational impact. Watson et al. (2017) explicitly note this research gap.

Have Students Enrolled in More Classes Due to Reduced Textbook Costs?

- No evidence exists that OER adoption increases student course-taking or credit enrollment—a critical metric if access is the true goal. The Florida Virtual Campus (2016) study found that while 47.6% of students reported taking fewer courses due to textbook costs, there is no published evidence that OER initiatives have actually increased course-taking rates. Institutions have not systematically tracked whether students enroll in more courses after OER adoption, despite this being obvious metric if access is genuine priority (Watson et al., 2017).
- The minimal research available suggests negligible enrollment effects. Colvard et al. (2018) found no significant difference in enrollment patterns before and after OER adoption at their institution. Course withdrawal rates show small effects: Clinton and Khan's (2019) meta-analysis found that withdrawal rates were slightly lower with OER (0.5-1 percentage-point decrease), but effect sizes were small and inconsistent across studies.
- The logic suggests limited enrollment impact because: Textbook costs are relatively small portion of total cost (2-4% of cost of attendance), students already strategize around textbook costs (buying used, sharing, using library reserves, going without), making one or two courses free doesn't enable taking additional courses when tuition for those courses is \$1,500-\$3,000 per course, and time constraints matter more than textbook costs for most students (working students can't take more credits regardless of material costs).

The Hidden Costs Students Actually Pay:

- While OER eliminates upfront textbook costs, students may pay far more in hidden costs and long-term consequences: Need to purchase additional materials or resources to supplement inadequate OER (study guides, test prep books, tutoring, online courses)—defeating "free" purpose, spend more time and cognitive effort teaching themselves content that should have been clearly explained, struggle academically due to poor materials (lower grades, need to retake courses, extended time to degree), lack preparation for advanced courses, internships, or jobs (requiring remediation or additional training), perform poorly on standardized professional exams (requiring expensive retakes), and miss out on

learning and intellectual development that quality materials would have provided—opportunity cost that affects career trajectory and earning potential.

- No research has comprehensively examined total costs to students. Watson et al. (2017) note that while OER efficacy research measures immediate course outcomes, no studies examine long-term impacts on "career preparedness, professional licensure exam pass rates, graduate school admissions and success, or lifetime earnings"—all potentially affected by foundational knowledge gaps from lower-quality OER.
- The false economy of "free but inadequate" materials may cost students tens of thousands more in career earnings, extended education, and lost opportunities than \$1,000 in textbook costs would have.

The Self-Teaching Burden Falls on Students:

- When OER materials are inadequate, students are left to teach themselves—seeking additional resources independently without guidance. They must: identify gaps in their understanding without framework for knowing what they don't know, search for supplementary materials without guidance on quality or appropriateness, teach themselves complex concepts using random YouTube videos, websites, or forums of unknown quality, spend extensive time (10-20+ hours per course) finding and learning from supplementary materials, and essentially pay tuition to self-teach while faculty who assigned inadequate materials may not acknowledge or address gaps.
- This creates enormous inequity: Students with prior knowledge, strong study skills, access to resources, and time can supplement successfully. First-generation students, working students, students without prior preparation, and those lacking time or resources struggle disproportionately. OER becomes another mechanism that advantages already-advantaged students while disadvantaging those it was meant to help.

Student Choice: Should Students Decide Between OER and Commercial Textbooks?

The Question Rarely Asked:

The topic of letting students choose between commercial and OER textbooks does not often come up in OER research, but it raises important issues such as autonomy, fairness, and learning styles. Right now, faculty or, sometimes, administrators pick the materials for entire courses, so every student uses the same resources. What if students could decide for themselves whether to buy a commercial textbook or use a free OER option?

Arguments for Student Choice:

- Respects student autonomy and diverse learning needs. Students are adults paying (often through loans) for education; they should have input into learning materials. Jhangiani and Jhangiani (2017) found that students value having options, and some students prefer print textbooks they can annotate, keep as references, and use without screens.
- Addresses equity concerns both ways. Student choice allows: students who cannot afford textbooks to use free OER without shame or stigma, AND students who can afford textbooks to purchase higher-quality materials if they judge OER inadequate. This avoids forcing all students to use potentially inferior materials to address the financial needs of some students.
- Enables natural quality comparison and feedback. If students could choose, their choices would provide valuable data about perceived quality, usefulness, and value. If most students chose commercial textbooks despite the cost, this would signal concerns about OER quality. If most chose OER, this would validate its adequacy.
- Reduces faculty burden and conflict. Rather than faculty having to choose between student finances and educational quality, students make their own cost-quality trade-offs based on individual circumstances, learning preferences, and priorities.

Arguments Against Student Choice (or Why This Isn't Implemented):

- Creates two-tiered learning experiences within same course. Students using different materials might have different depths of coverage and understanding, access different practice problems and resources, be prepared differently for exams, and experience different cognitive load and learning efficiency. Faculty would need to accommodate multiple materials, teach using multiple resources, and create assessments that are fair to both groups—significantly increasing the workload.
- Disadvantages students who choose free option. If OER is genuinely inferior (as research suggests it often is), students choosing it due to financial necessity would receive a worse education than wealthier peers—making the cost barrier visible and explicit within the classroom rather than hidden. This could create shame, resentment, and documented educational inequality.
- Complicates instruction and assessment significantly. Faculty couldn't reference specific textbook pages, examples, or problems if students have different materials. Homework from textbook problem sets wouldn't work if some students lack access. Lecture alignment with textbook would benefit only some students.

What Research Reveals About Student Preferences:

- Limited research exists because choice is rarely offered, but studies reveal interesting patterns. Jhangiani and Jhangiani (2017) found that when asked hypothetically: 67% of students said they would choose free OER over \$150 commercial textbook, but 42% of those same students rated OER quality lower than commercial textbooks they'd used previously, and 31% said they would purchase commercial textbook if they could afford it and believed it was significantly better quality.
- This suggests students make cost-quality trade-offs, not just cost minimization. Many students would pay for textbooks if they believed the quality difference was meaningful and they could afford it. The fact that textbooks cost \$200-\$400 (unconscionable amounts for many students) drives OER adoption, not that students believe free is always optimal.

What Would Happen If Student Choice Was Implemented?

Impacts on Universities:

- Would expose whether OER adoption is truly about student benefit or cost-cutting. If institutions claim OER is equal quality and students prefer it, offering choice should result in universal OER adoption. If students systematically choose commercial textbooks when given option, this reveals quality concerns and that institutions have been forcing inferior materials.
- Could increase administrative complexity and costs. Universities would need to: stock both OER and commercial textbooks (or provide access to both), manage two sets of materials in course reserves, accommodate different student materials in learning management systems, and track which students use which materials.
- Might reduce OER adoption if quality concerns are real. If significant numbers of students choose commercial textbooks, faculty might feel validated in quality concerns and push back against OER mandates, potentially derailing institutional OER initiatives and associated PR benefits.
- Would make cost-quality trade-offs explicit and potentially embarrassing. Media coverage of "students forced to choose between rent and textbooks" or "wealthy students get better education in same course" could create negative publicity institutions want to avoid.

Impacts on Professors:

- Would increase instructional workload significantly. Faculty would need to: accommodate multiple materials in course design, create assessments that are fair to students using different resources, field questions about both sets of materials, stay current with both OER and commercial options, and potentially teach the same content differently for different groups.

- Could validate faculty expertise and professional judgment. If student choices align with faculty quality assessments (students choosing commercial textbooks in courses where faculty believe OER is inadequate), this validates faculty professional judgment and could strengthen arguments for faculty autonomy.
- Would require communicating trade-offs to students. Faculty would need to advise students on which option to choose, explaining the strengths/weaknesses of each—time-consuming, potentially uncomfortable conversations about quality differences.

Why Student Choice Isn't Widely Implemented—The Uncomfortable Truth:

- Student choice would reveal quality differences institutions prefer to obscure. If OER is genuinely equal quality, why not offer choice? The fact that choice is rarely offered suggests that institutions know many students would choose commercial textbooks if they were affordable, revealing that "free" OER involves quality trade-offs that institutions don't want explicitly acknowledged.
- Universities benefit from mandatory OER because it reduces costs and maximizes PR claims. If only some students use OER, cost savings are reduced and headlines about "free textbooks for all students" become "free textbooks for some students," weakening marketing value.
- The current model allows institutions to control narrative and metrics. With student choice, students might choose differently than institutions prefer, complicating reporting and assessment.

When OER Works Well and Better Solutions

Conditions for OER Success:

- High-quality OER exists and can equal or exceed commercial textbooks under specific conditions. Research identifies success factors: adequate funding for professional development (Pitt et al., 2020), teams of discipline experts rather than individual authors (Allen et al., 2015), rigorous peer review processes (Jung et al., 2017), professional editing and design (Pawlyshyn et al., 2013), regular updating with sustainable funding models (Petrides et al., 2011), comprehensive ancillary materials (Delimont et al., 2016), faculty choice and autonomy rather than mandates (Belikov & Bodily, 2016), and adequate institutional support including course release time (Henson, 2017).
- OpenStax represents best-case scenario with \$20+ million in foundation funding, teams of experts, rigorous review, professional production, and ongoing maintenance—but most OER lacks these resources (Pitt, 2015).

Alternative Models That Balance Quality, Cost, and Choice:

- Inclusive Access Programs: Universities negotiate bulk rates with publishers for digital access to commercial textbooks, which are distributed to all students at reduced cost (\$25-\$50 per textbook instead of \$200-\$400) and charged as a course fee. Students can opt out if they prefer OER or used books. Research by Cummings et al. (2019) suggests this model: significantly reduces costs (60-75% savings), maintains quality (using vetted commercial materials), provides universal access (all students have materials day one), and reduces piracy and unauthorized sharing. However, concerns include: students are charged automatically without seeing the cost first (reduced transparency), can't opt out without extra effort (default enrollment), and publishers still profit while potentially undermining the used textbook market and OER alternatives.
- Curated OER Libraries with Quality Vetting: Some institutions (notably community college consortia) invest in vetting high-quality OER, providing faculty-curated collections reviewed for quality, and supplementing them with librarian and instructional design support. This addresses quality concerns while maintaining affordability.
- Affordable Textbook Grants/Vouchers: Some institutions provide all students grants or vouchers (\$200-\$500) to purchase required textbooks—commercial or OER, student choice—with institutional subsidy. This respects student autonomy,

ensures access regardless of financial need, and allows market forces to reveal quality preferences. However, this requires actual institutional financial commitment rather than cost-shifting.

What Would Actually Help:

- Regulate commercial textbook industry: Cap prices, prohibit forced new editions without substantial changes, unbundle access codes from textbooks, require affordable used textbook market access, and enforce anti-trust enforcement against publisher consolidation (Senack, 2016).
- Support high-quality OER development: Fund faculty course release time to create and maintain quality OER, provide professional development and editorial support, create rigorous peer review and quality vetting processes, establish sustainable funding for ongoing updates and maintenance, and support consortium approaches where multiple institutions collaborate (Allen et al., 2015; Pitt et al., 2020).
- Preserve faculty autonomy: Allow faculty to choose the best materials—OER, commercial, or hybrid—based on pedagogical needs and quality; provide stipends to faculty who create supplementary materials for OER; and respect professional judgment about which materials serve students best (Belikov & Bodily, 2016).
- Address root causes: Actually reduce tuition through state reinvestment in public higher education, reduce administrative bloat and executive compensation, prioritize instructional spending, and make higher education genuinely affordable (SHEEO, 2020).

The Bottom Line

Research on Open Educational Resources (OER) reveals a complex situation that requires thoughtful attention. The affordability crisis in commercial textbooks is well known, and there is evidence of publisher exploitation (GAO, 2013; Florida Virtual Campus, 2016). However, when institutions require OER adoption, it often seems driven more by public relations, cost savings, and political considerations than by evidence of better educational outcomes (Henson, 2017; Belikov & Bodily, 2016).

Several major repositories make up the OER landscape. OpenStax has reached 3.6 million students in over 14,000 courses (OpenStax, 2022). The BC Campus Open Collection has been adopted in more than 1,000 courses and saved students \$20 million (BCcampus, 2021). The Open Textbook Library offers over 950 textbooks, and LibreTexts has seen more than 1 billion accesses and over 12,000 course adoptions (Delmas et al., 2022). The AIM Open Textbook Initiative features 90 approved math textbooks. Even with these resources, OER is used in only about 8-10% of higher education courses (Seaman & Seaman, 2021). Adoption rates vary: 15-30% in lower-division general education courses at institutions with OER programs, 5-10% in upper-division courses, and 2-5% in specialized or graduate courses. Community colleges use OER more (12-15%) than four-year schools (6-9%), and states with funding see adoption rates two to three times higher than those without. While use has doubled from 3-4% in 2016 to 8-10% in 2021, most courses (over 90%) still use commercial textbooks. This may be due to a lack of high-quality OER for many courses, or to faculty concerns about whether available OER meets their needs.

OER is mostly used in lower-division general education courses, where commercial textbooks are the most expensive and attract the most attention. In contrast, commercial materials still dominate upper-division and specialized courses. This suggests that institutions may be focusing on reducing visible costs rather than on broad improvements in educational quality. Studies have found several quality issues: only 23% of OER materials are peer-reviewed (Jung et al., 2017), there are gaps in coverage and rigor (Bliss et al., 2013), 67% of faculty say there are not enough supplementary materials (Jung et al., 2017), 40-60% of OER is more than five years old (Pitt, 2015), and there are not enough practice problems in STEM subjects (Winitzky-Stephens & Pickavance, 2017).

Universities benefit from OER mandates by getting good publicity, cutting costs, and gaining political support, but they also shift unpaid work onto faculty. This unpaid labor is estimated at \$15,000- \$30,000 per course (Petrides et al., 2011), which can

undermine student learning. Many faculty feel pressured (63%; Belikov & Bodily, 2016), worry about quality (47%), and dislike the 20-60 hours of unpaid work needed for each course (Delimont et al., 2016). Research shows that student performance on basic measures is similar, but the studies have important limitations (Hilton, 2016). Some findings show worse results in higher-order thinking skills (Croteau, 2017; Grewe & Davis, 2017) and that 42% of students rate OER as lower quality (Jhangiani & Jhangiani, 2017). There is also no evidence that OER adoption increases enrollment, which would be important if access was the main goal (Watson et al., 2017).

There are better ways to address textbook costs, such as regulating commercial publishers (Senack, 2016), investing in high-quality OER with proper review and upkeep (like the OpenStax model with enough funding; Allen et al., 2015; Pitt et al., 2020), protecting faculty choice (Belikov & Bodily, 2016), trying inclusive access or subsidy models, and most importantly, increasing public investment to tackle the root causes (SHEEO, 2020). Students should have both affordable and high-quality materials, not one or the other. Current adoption rates—8-10% overall, mostly in lower-division courses, with over 90% of courses still using commercial materials—show that OER often acts as a symbolic fix, saving only 2-4% of total attendance costs, while tuition keeps rising (making up 50-70% of total costs and increasing by 20-35% in the same period). The "hidden cost of free" falls on students through lower educational quality, on faculty through unpaid work, and on society by producing graduates who are less prepared for their careers. This allows institutions to appear to be addressing affordability issues without making meaningful changes to their priorities.

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